Rod Cutting

Problem Statement

Given a rod of length 'n', we are asked to cut the rod and sell the pieces in a way that will maximize the profit. We are also given the price of every piece of length 'i' where '1 <= i <= n'.

Example:

Lengths: [1, 2, 3, 4, 5]
Prices: [2, 6, 7, 10, 13]
Rod Length: 5

Let's try different combinations of cutting the rod:

Five pieces of length 1 => 10 price
Two pieces of length 2 and one piece of length 1 => 14 price
One piece of length 3 and two pieces of length 1 => 11 price
One piece of length 3 and one piece of length 2 => 13 price
One piece of length 4 and one piece of length 1 => 12 price
One piece of length 5 => 13 price

This shows that we get the maximum price (14) by cutting the rod into two pieces of length '2' and one piece of length '1'.

Try it yourself

Try solving this question here:

